

Some definitions

$Subst := set (Var * Exp)$
 $Problem := set Equation$
 $Tuple := set (Subst * Problem)$
 $-\langle - \rangle : exp \rightarrow Subst \rightarrow exp$
 $dom : Subst \rightarrow set Var$
 $smatch : Tuple \rightarrow Tuple \rightarrow Prop$

smatch steps are classified into three

Congruence steps

$$\begin{aligned} \text{smatch_app_congr} : (s \approx t \approx s' \approx t') \in P \rightarrow \\ \text{smatch } (S, P) (S, P + (s \approx s') + (t \approx t') \setminus (st \approx s' t')) \end{aligned}$$

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Congruence steps

$$\begin{aligned} \textit{smatch_app_congr} : (s \ t =_? \ s' \ t') \in P \rightarrow \\ \textit{smatch} \ (S, P) \ (S, P + (s =_? \ s') + (t =_? \ t') \setminus (st =_? \ s' \ t')) \end{aligned}$$

σ -min steps

$$\begin{aligned} \textit{smatch_subst_comp} : (s[\sigma \circ \tau] =_? \ s'[\rho]) \in P \rightarrow \\ \textit{smatch} \ (S, P) \ (S, P + (s[\sigma][\tau] =_? \ s'[\rho]) \setminus (s[\sigma \circ \tau] =_? \ s'[\rho])) \end{aligned}$$

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Instantiation case

$$\begin{aligned} \textit{smatch_inst} : (s \approx X) \in P \rightarrow \\ X \notin \textit{lhs_vars} P \rightarrow \\ S' =_{\sigma_{\min}} S \gg \{(X, s)\} \rightarrow \\ \textit{smatch} (S, P) (S', (P \setminus (s \approx X)) \langle \{(X, s)\} \rangle) \end{aligned}$$

Preservation statement

Validity

$$\text{valid_tuple } (S, P) := \underline{(\text{dom } S) \cap (\text{vars_of } P) = \phi}$$

Matching solution

$$\text{match_sol } SI \ (S, P) := \frac{(s =_? t) \in P \rightarrow s =_{\sigma_{min}} t \langle SI \rangle \wedge \underline{\exists S', S \gg S' =_{\sigma_{min}} SI}}{}$$

Preservation theorem

$$\begin{aligned} \text{match_pres} : & \text{valid_tuple } T \rightarrow \\ & (\text{dom } SI) \cap (\text{lhvars_of } P) \rightarrow \\ & \text{smatch } T \ T' \rightarrow \\ & \text{match_sol } SI \ T' \rightarrow \\ & \text{match_sol } SI \ T \end{aligned}$$

Proof proceeds by case analysis on *smatch*

Inst case

$$H_1 : (\text{dom } S) \cap \text{vars_of } P = \{\}$$

$$H_2 : (\text{dom } SI) \cap \text{lhvars_of } P = \{\}$$

$$H_3 : X \notin \text{lhvars_of } P$$

$$H_4 : \text{match_sol } SI (S \gg \{(X, s)\}, (P \setminus (s, X)) \langle \{(X, s)\} \rangle)$$

match_sol SI P

Inst case (Unfolded *match_sol*)

$$H_1 : (\text{dom } S) \cap \text{vars_of } P = \{\}$$

$$H_2 : (\text{dom } SI) \cap \text{lhvars_of } P = \{\}$$

$$H_3 : X \notin \text{lhvars_of } P$$

$$H_4 : \forall s' t', (s' =_? t') \in (P \setminus (s, X)) \langle \{(X, s)\} \rangle \rightarrow s =_{\sigma_{\min}} t \langle SI \rangle$$

$$H_5 : \exists S', SI =_{\sigma_{\min}} (S \gg \{(X, s)\}) \gg S'$$

$$\forall s_0, t_0, (s_0 =_? t_0) \in P \rightarrow s_0 =_{\sigma_{\min}} t_0 \langle SI \rangle \quad \wedge$$

$$\exists S', SI =_{\sigma_{\min}} S \gg S'$$

Inst case continued..

$$H_1 : (\text{dom } S) \cap \text{vars_of } P = \{\}$$

$$H_2 : (\text{dom } SI) \cap \text{lhvars_of } P = \{\}$$

$$H_3 : X \notin \text{lhvars_of } P$$

$$H_4 : \forall s' \ t', (s' =_? t') \in (P \setminus (s, X)) \langle \{(X, s)\} \rangle \rightarrow s' =_{\sigma_{\min}} t' \langle SI \rangle$$

$$H_5 : \exists S', SI = (S \gg \{(X, s)\}) \gg S'$$

$$s_0, t_0 : \text{exp}$$

$$H_6 : (s_0 =_? t_0) \in P$$

$$s_0 =_{\sigma_{\min}} t_0 \langle SI \rangle$$

Inst case continued..

$$H_1 : (dom\ S) \cap vars_of\ P = \{\}$$

$$H_2 : (dom\ SI) \cap lhvars_of\ P = \{\}$$

$$H_3 : X \notin lhvars_of\ P$$

$$H_4 : \forall\ s'\ t', (s' =_? t') \in (P \setminus (s, X)) \langle \{(X, s)\} \rangle \rightarrow s' =_{\sigma_{min}} t' \langle SI \rangle$$

$$H_5 : \exists S', SI = (S \gg \{(X, s)\}) \gg S'$$

$$s_0, t_0 : \text{exp}$$

$$H_6 : (s_0 =_? t_0) \in P$$

$$s_0 =_{\sigma_{min}} t_0 \langle SI \rangle$$

Case analysis : $\{s_0 =_? t_0 = s =_? X\} + \{s_0 =_? t_0 \neq s =_? X\}$